Where to Predict in Retinal OCT? Anatomy-Guided Target Selection for I-JEPA

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Abstract

Self-supervised learning can reduce the dependence of retinal optical coherence tomography (OCT) models on expert labels, which are costly at scale and must capture subtle disease-related changes. Image-based Joint-Embedding Predictive Architecture (I-JEPA) learns by predicting target-region representations from visible context rather than reconstructing pixels. However, its uniform target placement does not account for the layered organization of retinal OCT, where retinal tissue occupies only part of each B-scan. We investigate anatomy-guided target selection using image intensity and predicted retinal segmentation while retaining the I-JEPA architecture and latent prediction objective. We compare tissue-directed rectangular targets, anatomy-shaped targets and coverage-constrained placement with uniform masking, and evaluate the learned representations through frozen-encoder glaucoma classification on FairVision, repeating the main comparison across further pretraining continuations under a registered analysis plan. In the original single runs, tissue-directed rectangle placement gave higher frozen-probe AUC than uniform masking, including with segmentation-free intensity guidance. The registered repeated continuations did not all finish before the results freeze, so they are reported descriptively and do not establish the direction of the effect. In the original runs, more anatomically specific strategies did not consistently provide further gains. Analysis of the delivered masks shows that target selection, target size and visible context are coupled. Our findings motivate anatomy-guided predictive learning for retinal OCT and better-controlled comparisons of what to predict and what to keep visible.

1 Introduction

Optical coherence tomography (OCT) reveals the layered structure of the retina and supports glaucoma assessment (Maetschke et al., 2019; Ran et al., 2019; Noury et al., 2022). Self-supervised pretraining supports medical imaging foundation models by learning reusable representations without requiring a diagnosis for every training scan (Azizi et al., 2021; Zhou et al., 2023; Qiu et al., 2024). The spatial organization of OCT makes the choice of pretraining task particularly relevant: retinal tissue forms a relatively narrow region within a B-scan, surrounded by vitreous and other background.

Image-based Joint-Embedding Predictive Architecture (I-JEPA) (Assran et al., 2023) learns by predicting hidden image representations rather than reconstructing hidden pixels. From the visible context and the target positions, a predictor estimates the representations of a target encoder updated as an exponential moving average (EMA) of the context encoder; no labels are needed. Unlike masked autoencoders (He et al., 2022; Xie et al., 2022), I-JEPA does not require the predictor to reproduce each pixel.

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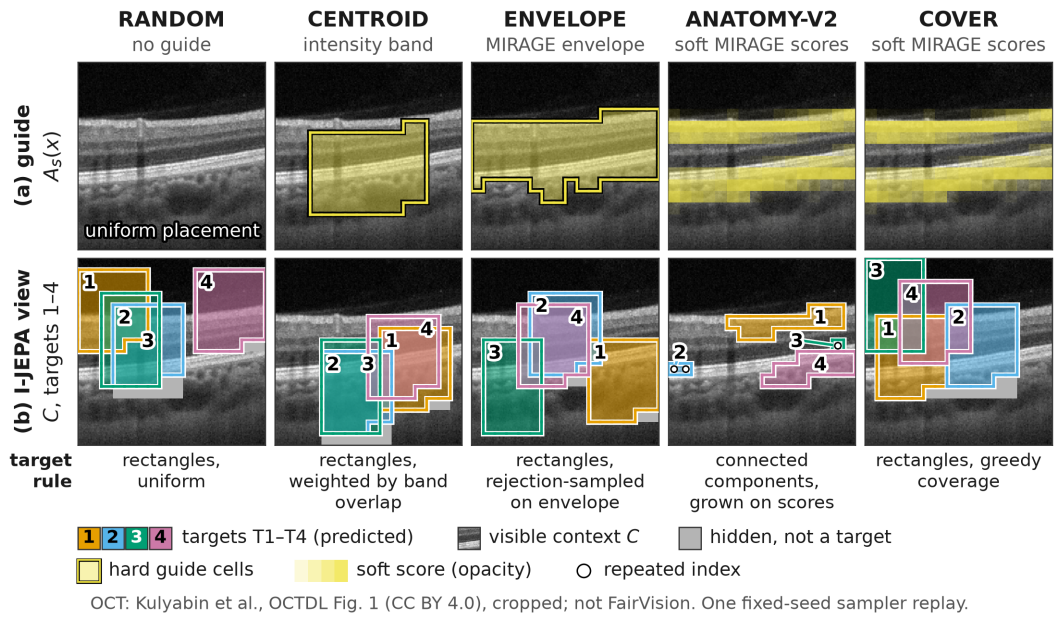


Figure 1: The five target-selection strategies on one external OCT example, not FairVision data: an annotation-free B-scan crop from Kulyabin et al., OCTDL, Fig. 1 (arXiv:2312.08255v3), adapted (overlays added) under CC BY 4.0 and downsampled to 256 by 256 pixels without contrast enhancement (licence: `https://creativecommons.org/licenses/by/4.0/`). (a) Placement guide $A_s(x)$ (Section 3); MIRAGE was not trained on OCTDL, and ANATOMY-V1 is not shown. (b) One I-JEPA view: visible context C, target groups T_m (colour and number mark identity, not relevance), hidden non-targets (grey) and indices repeated within a target (circles); the target rule follows Table 1. RANDOM targets include much background, the guided rectangles cluster on the tissue band and overlap, and ANATOMY-V2 targets are thin tissue-shaped components. Exact indices of one fixed-seed replay of each trained sampler on this image (matched block sizes); illustrative only, not a segmentation, training batch, evaluation sample or performance comparison.

Our interest in anatomical guidance also arose from exploratory feature-occlusion maps of glaucoma classifiers initialized from uniform-masking I-JEPA and then fine-tuned with diagnosis labels, in which selected maps suggested prediction sensitivity near retinal structures (Appendix F).

Standard I-JEPA places rectangular targets uniformly across the image. In retinal OCT, many such targets include little retinal tissue. This raises our central question: *can I-JEPA benefit from selecting prediction targets according to retinal anatomy rather than uniformly across the image?* Figure 1 illustrates the change in target placement. Semantic and anatomy-guided masked modelling provide positive motivation for this question (Li et al., 2022; Kakogeorgiou et al., 2022; Li et al., 2024; Wang et al., 2025). However, selecting a target also removes information from the context. The useful learning task depends on both what is predicted and what remains visible.

We investigate this question through five masking families: uniform RANDOM, the segmentation-free CENTROID, segmentation-guided ENVELOPE, anatomy-shaped targets (two versions) and coverage-constrained COVER, the last three guided by MIRAGE retinal segmentation (Morano et al., 2025). They vary target location, shape or coverage within the unchanged I-JEPA architecture and latent objective. We evaluate the resulting B-scan representations by pooling them across an OCT volume and fitting a linear glaucoma classifier.

In the original runs, each a single pretraining continuation per strategy from a shared epoch-25 checkpoint, tissue-directed rectangle placement gave higher frozen-probe AUC than uniform masking. More anatomically specific targeting did not improve on these simpler strategies: ANATOMY-V2 and COVER scored below ENVELOPE at every matched epoch from epoch 50 on. To understand what the comparisons test, we measure the target and context patches that actually reach the model. At the training batch size, CENTROID leaves 4.0 percentage points more of the patch grid visible to the context encoder than RANDOM, and ENVELOPE leaves 1.6 points more (Table 3). The strategies

also differ in target size and overlap, and batching shortens COVER's targets after they are scored, so its delivered coverage differs from the intended one. Target location and visible context therefore change together, and we present comparisons of implemented masking strategies, not controlled ablations of target location. The registered repeated continuations did not all finish before the results freeze, so they are reported descriptively.

Contributions.

- A segmentation-free target guide, CENTROID, that steers I-JEPA target placement toward an intensity-centroid band in each B-scan, compared with segmentation-guided and uniform target selection on retinal OCT.
- Frozen-probe results showing gains from tissue-directed rectangle placement in the original runs, with the registered repeated continuations reported descriptively.
- An analysis of the target and context masks delivered at the training batch size, which shows how target placement, target size and visible context change together and identifies constraints for better-controlled guided-masking studies.

2 Related work

Masked image modelling and JEPAs. Masked autoencoders reconstruct pixels (He et al., 2022; Xie et al., 2022), BEiT predicts discrete visual tokens (Bao et al., 2022) and iBOT distils teacher outputs at masked patch positions (Zhou et al., 2022), whereas data2vec and I-JEPA predict latent representations (Baevski et al., 2022; Assran et al., 2023). Masked prediction also extends to video and volumetric data (Bardes et al., 2024; Chen et al., 2023b; Taleb et al., 2020), and JEPA formulations differ in objectives and prediction structure (Balestriero and LeCun, 2025; He et al., 2025). We retain the I-JEPA objective and study target selection.

Region selection and prediction structure. DSeq-JEPA combines attention-derived region selection with sequential prediction (He et al., 2025). Its highest-ranked region supplies context for subsequent predictions, and its ablation favours the joint design over region selection alone. We instead use retinal anatomy with parallel I-JEPA prediction, where the guidance map, selected targets and available context jointly define the learning task. I-JEPA's own masking analysis emphasizes both sufficiently large targets and informative context (Assran et al., 2023).

Medical region-conditioned prediction. US-JEPA uses a frozen ultrasound teacher, and its region-conditioned variant restricts both context and target sampling to valid ultrasound content (Radhachandran et al., 2026). We instead study how intensity-based and retinal-segmentation guidance change target selection within the EMA-based I-JEPA framework for OCT, without sequential prediction or a new teacher architecture. The contribution is this OCT comparison and analysis of the resulting prediction tasks, not the general idea of informed masking.

Informed masking. ADIOS learns adversarial masks (Shi et al., 2022); SemMAE uses semantic parts (Li et al., 2022); AttMask selects high-attention patches (Kakogeorgiou et al., 2022); and HPM (hard patches mining) and AutoMAE adapt masking to prediction difficulty (Wang et al., 2023; Chen et al., 2023a). Medical approaches include AnatoMask, AnatomyMAE, MSMAE and VasoMIM (Li et al., 2024; Ceballos Arroyo et al., 2026; Mao et al., 2023; Huang et al., 2026). These studies motivate informed masking, but their objectives, guidance sources and context construction differ from anatomy-guided target selection under latent prediction on retinal OCT.

Retinal foundation models and OCT. RETFound (Zhou et al., 2023), VisionFM (Qiu et al., 2024), UrFound (Yu et al., 2024) and OCTCube (Liu et al., 2026) pretrain on large retinal corpora. OCT-specific masked modelling is explored by Pissas et al. (2024). MIRAGE (Morano et al., 2025) provides multimodal OCT segmentation and supplies the anatomical guidance used by several of our strategies, and deep glaucoma detection from OCT is established (Maetschke et al., 2019; Ran et al., 2019; Noury et al., 2022). FairVision provides the glaucoma-labelled OCT data used for our downstream comparison (Luo et al., 2024).

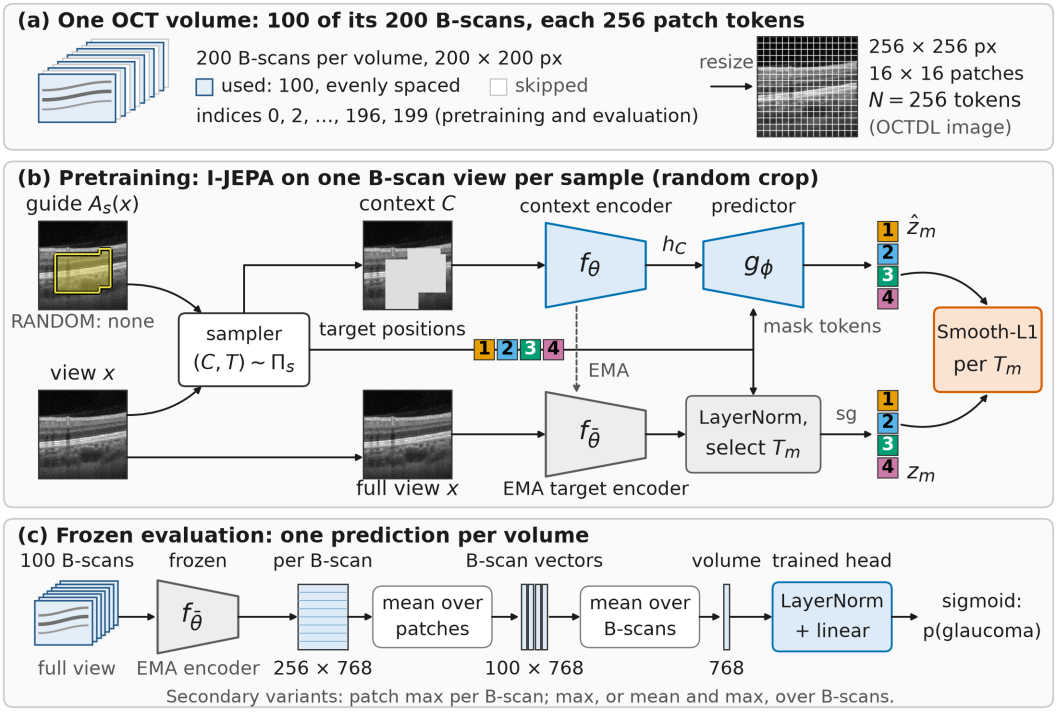


Figure 2: From an OCT volume to I-JEPA pretraining and volume-level evaluation. (a) Each FairVision volume has 200 B-scans; pretraining and evaluation use the same 100 evenly spaced ones, resized to 256 by 256 pixels ($N = 256$ patches of 16 by 16 pixels). (b) Each B-scan is a separate sample (random resized crop). Strategies differ in the sampler of context C (grey: withheld tokens, targets included) and targets T_m (uniform or from $A_s(x)$; CENTROID's masks of Figure 1); encoders, predictor and loss are shared. The predictor sees h_C and mask tokens at the target positions; the EMA target encoder's normalized outputs at T_m are the stop-gradient targets of the Smooth-L1 loss (Equation 3). (c) The frozen EMA encoder sees uncropped B-scans; patch features are averaged per B-scan, then across the volume, before a LayerNorm–linear head. Stacks are schematic; the B-scan is the OCTDL crop.

3 Background and target-selection framework

The architecture and latent prediction loss are shared; differences in guide sources, target size and final context are part of the strategy comparisons.

3.1 I-JEPA preliminaries

Let x be a B-scan partitioned into N patches, and let C denote the indices of the patches supplied to the context encoder. Target group T_m contains K_m patch indices, with $m = 1, \ldots, M$. We treat T_m as an index list because some implementations repeat indices when equalizing target lengths. The context excludes the union of target positions. Writing p_C and p_{T_m} for the corresponding positional embeddings, the context and prediction paths are

$$h_C = f_\theta(x; C), \qquad \widehat{z}_m = g_\phi(h_C, p_C, p_{T_m}). \tag{1}$$

Here $f_\theta(x; C)$ embeds only the selected patches at their original image positions. The predictor projects h_C to its own feature dimension and adds its own positional embeddings p_C; target queries combine a learned mask token with p_{T_m}. The target encoder sees the full B-scan. Its output is normalized per patch before selecting the target positions:

$$z_m = \mathrm{sg}([\mathrm{LN}(f_{\bar{\theta}}(x))]_{T_m}), \qquad \bar{\theta} \leftarrow \mu\bar{\theta} + (1-\mu)\theta. \tag{2}$$

The operator sg stops gradients, LN is LayerNorm across feature dimensions, and μ is the EMA coefficient. The predictor uses the same context for each target group; predictions are not conditioned on earlier target predictions.

Each target and prediction is a $K_m \times D$ matrix. For feature dimension D, the implementation minimizes the mean Smooth-L1 loss over target tokens and feature coordinates:

$$\mathcal{L} = \frac{1}{D \sum_{m=1}^{M} K_m} \sum_{m=1}^{M} \sum_{k=1}^{K_m} \sum_{d=1}^{D} \rho(\widehat{z}_{m,k,d} - z_{m,k,d}) \,. \tag{3}$$

The Smooth-L1 function ρ uses the standard unit transition between its quadratic and linear regions. We call the $\sum_m K_m$ supervised target tokens *loss slots*; repeated or overlapping indices add loss slots, and the mean is taken over all loss slots of a batch (Appendix D). Figure 2 shows the pretraining and evaluation paths.

3.2 Guided target selection

Background patches above and below the retina often have low intensity variation, whereas tissue patches carry the layered structure. Guided strategies direct targets toward tissue so that prediction depends more on it. However, background-position embeddings can incorporate tissue information through attention; visually uniform pixels need not imply uninformative targets or context. For strategy s, let $A_s(x)$ denote its patch-grid guidance: an intensity-derived band for CENTROID, a repaired segmentation envelope for ENVELOPE, or soft retinal-class score maps for the anatomy-shaped and coverage strategies. CENTROID is an intensity-guided tissue-localization baseline, not a segmentation model. We write the joint masking procedure as

$$(C, T) \sim \Pi_s(\cdot \mid A_s(x)), \qquad T = (T_1, \ldots, T_M). \tag{4}$$

Π_s includes target placement and context construction, including batch-dependent token equalization. A candidate context block is sampled, target positions are excluded, and token lists are shortened as needed for batching. Only the remaining context tokens enter f_θ. Consequently, tissue outside the target union need not be visible to the encoder. We report *unique targets* (cells in the target union), *tissue hidden* (share of tissue cells inside that union), *purity* (share of target cells on tissue) and *context* (share of the grid supplied to the context encoder); the target union and the context need not be complementary.

3.3 Compared masking strategies

All policies request $M{=}4$ target blocks and one context block (Table 1, Figure 1). Rectangle strategies share nominal scale and aspect-ratio ranges, but placement and collation can change unique targets and encoder-visible context.

RANDOM (baseline). Stock I-JEPA multiblock. Four rectangles placed uniformly at random anywhere in the grid. No image content is consulted.

ENVELOPE (random-within-retina). The same nominal rectangle count, scale and aspect-ratio distributions as RANDOM, but rejection-sampled onto the MIRAGE-derived repaired retinal envelope (Morano et al., 2025). The guide model is MIRAGE-Large fine-tuned for layer segmentation on GOALS circumpapillary B-scans (Fang et al., 2022). Its held-out GOALS foreground Dice of 0.925 is in-domain context only, not an accuracy on FairVision, which has no layer annotations and a different scan protocol. The predicted inner-retina and choroid union is repaired across the unlabelled middle-retina gap in native label space, then transformed jointly with the image. This changes the placement policy, not target shape. Overlap and delivered context are consequences of that intervention, not independently matched controls.

CENTROID (segmentation-free). Let $\bar{x}_{ij}$ be the mean intensity of the patch in row i and column j, minus the smallest patch mean in the image. The band centre in column j is the intensity-weighted row centroid

$$c_j = \textstyle\sum_i i\, \bar{x}_{ij} \big/ \sum_i \bar{x}_{ij}, \tag{5}$$

averaged over neighbouring columns. A band of fixed height centred on c_j across the central columns (Appendix A) weights the placement of four rectangular targets; it is not itself the target shape. No segmentation model or annotation of any kind is used.

ANATOMY-V1 / ANATOMY-V2. Connected components grown on soft MIRAGE retinal-class scores from a frozen feature adapter (Appendix A), so target *shape* follows tissue. Each

Table 1: Compared strategies. Variable-length rectangle targets are shortened to the shortest target in the batch (Appendix D). Delivered target and context budgets are reported in Table 3.

policy	guide $A_s(x)$	target shape	placement	target length
RANDOM	none	rectangles	uniform	variable
ENVELOPE	hard MIRAGE envelope	rectangles	rejection-sampled on envelope	variable
CENTROID	intensity-centroid band	rectangles	weighted by band overlap	variable
ANATOMY-V1/V2	soft MIRAGE scores	connected components	grown on scores	fixed (K)
COVER	soft MIRAGE scores	rectangles	greedy coverage	variable

component is resampled to exactly $K{=}16$ indices, subsampled when larger and topped up with repeated indices when smaller (circles in Figure 1). v2 also bridges diagonal adjacencies.

COVER f. Targets chosen greedily to cover anatomy while leaving a fraction f of supported soft guide mass outside the target union. The trained configuration uses $f{=}0.21$. This condition does not guarantee the same fraction survives final encoder-context collation, and post-placement target truncation changes the intended coverage (Appendix D).

4 Experimental setup

Data. We use FairVision OCT volumes with binary glaucoma labels (Luo et al., 2024) and the fixed released splits. Training and validation contain 6,000 and 1,000 volumes, respectively. Each volume contributes 100 of its 200 B-scans, evenly spaced and resized to $256{\times}256$ (Figure 2a). The test split contains 3000 volumes (1466 with glaucoma). Diagnosis labels are used only for downstream evaluation, not JEPA pretraining. Demographic attributes are not model inputs.

Pretraining. Patch-level I-JEPA, ViT-Base, $256{\times}256$ crops, patch size 16, predictor depth 6, EMA target encoder. The original continuations fork from a shared epoch-25 checkpoint of uniform-masking pretraining, with matched nominal optimizer, learning-rate and weight-decay schedules and effective batch size (512). The probability of guided placement rises linearly to full strength between epochs 25 and 30. The lineage of the early ANATOMY-V1 continuation is described in Appendix A. The encoder processes B-scans independently, without attention across B-scans. Table 1 and Appendix A list the remaining differences between strategies.

RANDOM, CENTROID, ENVELOPE and COVER reach epoch 100. ANATOMY-V2 reaches epoch 75; its epoch-100 result is unmeasured. We compare strategies at matched pretraining epochs, with epoch 50 as the common endpoint for the five main strategies. Excluded continuations are listed separately in Appendix A.

Replication and controls. The original comparison has one continuation per strategy. A replication analysis was registered before any new run was evaluated (Appendix B). It adds three further continuations each of RANDOM, CENTROID and ENVELOPE from the same epoch-25 checkpoint, with seeds that change data order, crops and mask draws, stopped at epoch 50, and a control that places targets uniformly with CENTROID's per-image target and context budgets. These are repeated continuations from a shared checkpoint, not independent initializations. They are evaluated on the same test split as the original runs. Test predictions were stored sealed and scored once, after the results freeze, and all run decisions used validation AUC. The third continuation of each strategy was added by a registered amendment before any new result. Six of the ten registered continuations finished before the results freeze. ENVELOPE seed B finished after the freeze and is reported descriptively. The seed-C runs did not produce an epoch-50 result before the freeze and are reported as not completed.

Evaluation. The EMA target encoder is frozen and encodes each B-scan without cropping. We average patch features within each B-scan, then average across the volume's B-scans and train a LayerNorm–linear head (Figure 2c). Each head uses a 50-epoch budget and is selected by validation AUC. We report test AUC with percentile bootstrap intervals and paired differences (10,000 class-stratified resamples, fitted heads fixed), together with DeLong tests (DeLong et al., 1988). These

Table 2: Frozen mean-pooled linear probe, FairVision test (3000 volumes, 1466 with glaucoma). The reference ancestor is epoch 25 (AUC 0.8487). Δ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries the original-run comparisons; the lower block is fp32, each Δ taken against an fp32 null re-probed at the same epoch. The measured fp16/fp32 gap is at most 2×10^{-4} (Appendix H). The last row is a value reported by FairVision for a different, fully supervised protocol. Bold Δ: the paired bootstrap interval excludes zero.

		epoch 50		epoch 75		epoch 100	
strategy	guidance signal	AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	none (null)	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	segmenter (location only)	0.8761	$\mathbf{+0.0120}$	0.8803	$\mathbf{+0.0080}$	0.8807	$\mathbf{+0.0062}$
CENTROID	intensity centroid, no segmenter	0.8740	$\mathbf{+0.0099}$	0.8836	$\mathbf{+0.0113}$	0.8855	$\mathbf{+0.0109}$
ANATOMY-V2	segmenter (target shape)	0.8654	$+0.0013$	0.8612	$\mathbf{-0.0111}$	—	—
COVER f=0.21	segmenter (coverage)	0.8643	$+0.0002$	0.8639	$\mathbf{-0.0084}$	0.8577	$\mathbf{-0.0168}$
reported: supervised 3D ResNet (Luo et al., 2024)		0.8649 (trained end to end on all B-scans; not this probe)					

intervals describe test-case sampling, not variation across pretraining runs. Probe precision is described in Appendix H.

5 Results and analysis

These are strategy comparisons, not factorial ablations: each original strategy has one pretraining continuation and may change several aspects of the prediction task. Section 5.3 adds the registered repeated continuations.

5.1 Comparison of target-selection strategies

Tissue-directed rectangles. At the common endpoint, epoch 50, ENVELOPE and the segmentation-free CENTROID exceed uniform masking by $+0.0120$ AUC (CI $[+0.0068, +0.0173]$) and $+0.0099$ (CI $[+0.0051, +0.0149]$). Their paired differences from RANDOM are positive at all three evaluated epochs, from $+0.0062$ to $+0.0120$ for ENVELOPE and from $+0.0099$ to $+0.0113$ for CENTROID (Figure 3; multiple-comparison adjustment in Appendix G). These gains favour tissue-directed target placement in the original runs. Both strategies also leave more of the grid visible to the context encoder than RANDOM (Table 3), so these comparisons do not separate target location from visible context. The gains do not require targets to contain only retinal tissue: the large rectangles also include background.

Anatomy-shaped targets. At the matched epoch 50, ANATOMY-V2 has lower AUC than ENVELOPE ($\Delta = -0.0107$; CI $[-0.0167, -0.0046]$). At epoch 50, the intervals for both ANATOMY-V2 and COVER versus RANDOM include zero.

Coverage-constrained placement. COVER is close to RANDOM at epoch 50 ($+0.0002$, CI $[-0.0050, +0.0053]$) but falls behind by epoch 100 (-0.0168, CI $[-0.0233, -0.0105]$), as its own AUC declines after epoch 73 while RANDOM keeps improving (Figure 3). Batching shortens its targets after placement, so the trained policy is not a controlled higher-coverage counterpart to ENVELOPE (Appendix D).

In the original single runs, the CENTROID–ENVELOPE difference at the final evaluated epoch is $+0.0047$ (CI $[+0.0004, +0.0091]$). The paired CENTROID–ENVELOPE difference intervals include zero at epochs 50 and 75. A simple intensity-based guide is therefore a useful comparator for segmentation-guided methods.

5.2 Analysis of target and context masks

Table 3 reports the masks that reach the model. At the training batch size, CENTROID leaves 4.0 points more of the grid visible than RANDOM (pooled 95% interval over batches 3.6–4.4), and ENVELOPE 1.6 points more (1.1–2.1). With loss slots equal across the rectangle strategies, CENTROID's

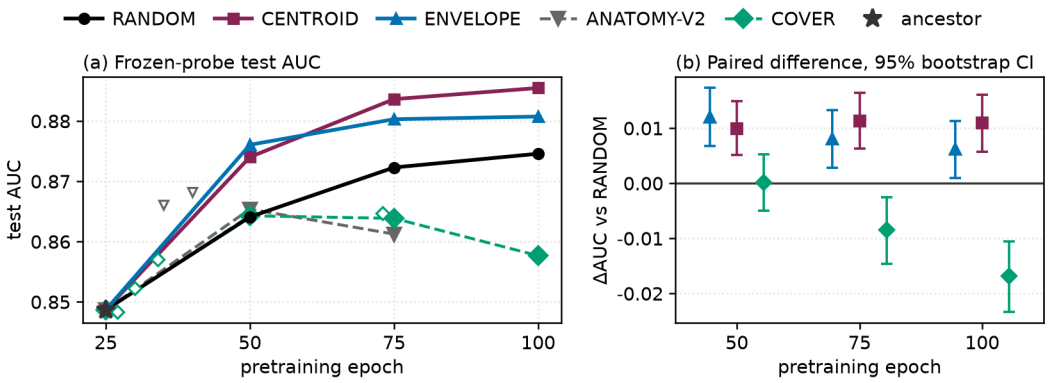


Figure 3: **(a)** Frozen-probe test AUC against pretraining epoch; note the truncated AUC axis. Lines join the shared epoch-25 ancestor (star) to epochs 50, 75 and 100; open markers are other checkpoints, not joined. Solid lines are fp16 probes and dashed lines fp32 probes; each Δ uses a precision-matched RANDOM (fp16/fp32 gap at most 2×10^{-4}, Appendix H). **(b)** Paired differences from RANDOM on identical test cases, 10,000-resample percentile bootstrap intervals for the original runs (seed spread: Figure 4). Every ENVELOPE and CENTROID interval excludes zero; COVER straddles zero at epoch 50 and is negative thereafter.

Table 3: Masks delivered at the training batch size of 64, measured by replaying the sampler code in each strategy's trained configuration on 6,400 fixed training views. The masks used during the original training were not recorded. Values are means over 3 sampler seeds of 100 batches each, with crops and block sizes paired across strategies. The context column also gives the across-seed s.d. Tissue is a thresholded MIRAGE-derived proxy, not ground-truth annotation. Column definitions, and the values reported in the review version, are in Appendix A. AUC is at the matched epoch 50 (fp32 probes for COVER and ANATOMY-V2).

strategy	tissue hidden	purity	unique targets	loss slots	context (% of grid)	AUC (ep. 50)
RANDOM	53.2%	30.9%	113.1	160.2	27.5 ± 0.1	0.8641
CENTROID	59.7%	38.2%	102.5	160.2	31.5 ± 0.4	0.8740
ENVELOPE	70.7%	42.1%	110.2	160.2	29.2 ± 0.5	**0.8761**
COVER	73.3%	40.1%	120.1	160.2	25.2 ± 0.4	0.8643
ANATOMY-V2	79.7%	**96.9%**	54.0	64.0	63.3 ± 0.4	0.8654

rectangles overlap more and cover fewer unique targets than RANDOM's (Figure 1). ENVELOPE differs least from RANDOM in target and context budgets but has the highest tissue purity among the rectangle strategies. These context differences are smaller than in the review version, which used one unpaired replay; they keep their sign for CENTROID and ENVELOPE, whereas the COVER difference reverses (Appendix A).

ANATOMY-V2 has high tissue purity but a smaller target union and more total context than the rectangle strategies. Its fixed K=16 indices per target give $4 \times 16 = 64$ loss slots, compared with 160.2 for each rectangle strategy, an explicit target-length choice rather than a property of irregular shapes. Appendix A gives the measurement protocol and the per-image geometry before batching, and Appendix E audits the coverage strategy in complete training-sized batches.

5.3 Replication across continuations and matched-budget control

Repeated continuations. No registered label is assigned because fewer than two seed indices finished all three strategies before the results freeze. The completed runs are reported descriptively in Table 4. Including the original runs, which used earlier code and partly different hardware, the mean differences are $+0.0104$ for CENTROID and $+0.0103$ for ENVELOPE (descriptive). Seed index B did not finish all three strategies before the freeze, so its runs are descriptive.

Table 4: Registered replication at epoch 50: test AUC, mean over five probe-head seeds (fp32 probes, head-seed s.d. at most 0.0011). Original encoders are re-probed with the same protocol (within 0.0011 of Table 2). Mean and s.d. are over the new seeds in the registered analysis. Δ is the paired difference from RANDOM of the same seed, descriptive for the original row. *Seed index incomplete at the freeze, not in the analysis. Intervals: Table 7.

	RANDOM	CENTROID	ENVELOPE	Δ CENTROID	Δ ENVELOPE
original (re-probed)	0.8641	0.8752	0.8755	+0.0111	+0.0114
seed A	0.8624	0.8721	0.8716	+0.0096	+0.0092
seed B*	0.8635	0.8754	0.8710	+0.0120	+0.0075
mean (new seeds)	0.8624	0.8721	0.8716	+0.0096	+0.0092
s.d. (new seeds)	—	—	—		

Table 5: Matched-budget control at epoch 50, seed A, one continuation per arm. Δ is the paired difference from RANDOM with a test-case interval. Budgets are per image at the training batch size: unique targets (U), loss slots (L), duplicates (D) and context tokens (C). RANDOM and CENTROID come from the audit of Table 3, RANDOM-CB comes from its run's mask statistics (3,740 batches, mean hidden-area residual 0.00) and RANDOM-CB equals its CENTROID shadow per image (0 mismatches in 960 image draws). Purity uses a hard MIRAGE-envelope tissue proxy (190 views), not comparable with Table 3.

policy	test AUC	Δ (interval)	U	L	D	C	purity (%)
RANDOM	0.8624	—	113.1	160.2	47.1	70.5	36.0
RANDOM-CB	0.8650	+0.0026 [−0.0015, +0.0066]	102.2	159.2	57.1	79.5	36.7
CENTROID	0.8721	+0.0096 [+0.0052, +0.0142]	102.5	160.2	57.7	80.8	46.0
CENTROID − RANDOM-CB		+0.0070 [+0.0030, +0.0111]	placement at matched budgets				

Matched-budget control. RANDOM-CB receives CENTROID's per-image unique targets, loss slots and visible context but places its targets uniformly. Its difference from CENTROID therefore reflects placement at equal budgets, and its difference from RANDOM reflects the change of budget. ENVELOPE is not budget-matched in this design. With these budgets, RANDOM-CB differs from RANDOM of the same seed by +0.0026. The placement difference, CENTROID minus RANDOM-CB, is +0.0070 (descriptive). No registered description is assigned because fewer than two seed indices finished all three strategies before the results freeze. The comparison is a single continuation per arm, and the two policies differ only modestly in target tissue purity (46.0% versus 36.7%), so a null placement difference would be weak evidence that placement does not matter.

Pooling. The registered analysis also probes every new and re-probed encoder with the three secondary poolings of Figure 2c; the patch maximum within B-scans may keep localized responses that the patch mean attenuates. Across the alternative poolings, the mean difference from RANDOM over the new continuations ranges from +0.0142 to +0.0188 for CENTROID and from +0.0069 to +0.0184 for ENVELOPE (Table 8). Where a label can be assigned, it is the same as with the primary pooling. These analyses are secondary and descriptive.

6 Discussion and limitations

The rectangle-policy results of the original runs motivate anatomy-guided target selection, but the registered repeated continuations did not all finish, so the direction of these gains across runs is not established. Masking studies should report the budgets that reach the encoder at the training batch size, not only the requested ones (Section 5.2).

Experimental limits. Each original strategy has one pretraining continuation, so its test-case confidence intervals do not establish a ranking over retrainings (Bouthillier et al., 2021). The registered replication adds continuations from the same epoch-25 checkpoint. It measures variation across continuations, not across independent initializations. Only one further continuation per strategy entered the registered analysis, the endpoint is epoch 50 only, and the per-seed intervals describe test cases, not retraining. The registered replication did not finish before the results freeze, so its completed runs are descriptive. The matched-budget control is a single continuation per arm. The

early anatomy-shaped result has incomplete continuation documentation, and ANATOMY-V2 was stopped after inspecting its epoch-75 deficit. Its later trajectory is unknown. Probe optimization and initialization can also affect frozen-probe comparisons; the reported fixed-budget protocol is not a measurement of the best achievable classifier for each encoder. External foundation models such as RETFound under the same frozen probe, other self-supervised objectives such as masked image modelling, and target-selection rules derived from a formal objective are left for future work.

The study uses one dataset and one downstream task. Strategies, checkpoints and analyses were chosen after repeated inspection of the same test split, although probe heads were fitted on training data and selected on validation data. Reported intervals are therefore descriptive rather than confirmatory. The replication uses the same test split with a plan registered before any new run was evaluated, which limits but does not remove this reuse. Validation-split differences are reported alongside every per-seed test difference (Appendix B). FairVision has no layer annotations, so guide accuracy cannot be measured on it. On 300 stratified training volumes, the CENTROID band disagreed grossly with the MIRAGE envelope in 4.8% of B-scans, more often in glaucoma and low-signal scans, and 3.8% of envelopes raised a plausibility flag (Appendix C). These are agreement statistics between two automatic guides, not anatomical accuracy. Only the specified coverage setting was pretrained, so we identify neither an optimal coverage level nor a mechanism for its AUC decline.

Broader impact and ethics. This study uses a public, de-identified retinal dataset under its release terms and involves no new recruitment or data collection, and no model reported here is clinically validated or intended for deployment (Appendix A).

6.1 Future work

An immediate priority is a controlled test of coverage on the masks delivered to the model. COVER should score post-equalization targets and verify retinal tissue in the final encoder context, rather than relying on coverage or visibility before collation (Appendices D and E). Any correction defines a new training configuration whose downstream effect remains untested. Comparisons should hold the guide product, delivered target-token and loss-slot budgets, and total visible context fixed where feasible, while varying placement and coverage. Remaining differences in visible tissue should be measured explicitly. Controlled ablations should also vary the tissue/background composition of targets and context at fixed token budgets.

Future replication should use complete preregistered seed sets and independently initialized pretraining runs. Evaluation on an untouched cohort, with fixed checkpoint, pooling and analysis rules, would strengthen evidence after the test-set reuse disclosed here. These future experiments do not change the current registration's treatment of late or incomplete runs.

Representative layer-annotated OCT scans are needed to measure guide accuracy, especially under low signal and pathology. Transfer to other datasets and spatial tasks such as segmentation would test whether the learned features retain information obscured by volume-level pooling. External models and other self-supervised objectives should be compared under the same probe protocol. Together, these studies would test the trade-off between anatomical target coverage and informative visible context.

7 Conclusion

We investigated where I-JEPA should predict in retinal OCT. Tissue-directed rectangle placement improved frozen-probe glaucoma AUC in the original runs, including a simple intensity-centroid strategy. More anatomically specific targeting did not consistently improve these results, and the implementations differ in more than target location, including the context they leave visible. The registered repeated continuations did not all finish before the results freeze. Comparisons that match guidance, target sizes and visible context across independent pretraining runs remain the next step.

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A Additional experimental details

Pretraining and linear evaluation. The input crop is 256×256, with patch size 16 and a ViT-Base encoder. The predictor has depth 6 and hidden dimension 384. Pretraining uses AdamW with peak learning rate 0.00025, a five-epoch warmup, final learning rate 0.000001, weight decay scheduled from 0.04 to 0.4, and EMA coefficient from 0.996 to 1.0. The effective batch size is 512, formed from batches of 64 with gradient accumulation. Rectangle target scales are 0.15–0.20 of image

area, aspect ratios 0.75–1.5, and context scales 0.85–1.0. Random resized crops use area scale 0.3–1.0. The guidance probability increases over the epoch-25 to epoch-30 continuation window; fixed-length anatomy targets use $K = 16$ throughout. The frozen linear head uses AdamW, learning rate 0.0004, weight decay 0.05, batch size 256, seed 42, and a 50-epoch budget with validation-AUC selection and patience 15. Paired analyses rely on the documented shared split and deterministic loader ordering; historical prediction files lack independent case-ID manifests.

B-scan selection and pooling. Each FairVision volume stores 200 B-scans of 200×200 pixels. Pretraining and evaluation use the same 100 B-scans per volume, with indices $\lfloor 199k/99 \rfloor$ for $k = 0, \ldots, 99$ (0, 2, …, 196, 199), each resized bilinearly to 256×256 pixels. Pretraining samples every B-scan independently as a random resized crop. Evaluation encodes the full B-scans, averages the 256 patch features of each B-scan and then the 100 B-scan vectors (dimension 768) before the LayerNorm–linear head. The registered secondary variants use the patch maximum within B-scans, the maximum across B-scans, or the concatenated mean and maximum across B-scans.

Guide sources. ENVELOPE uses the MIRAGE-derived repaired hard retinal envelope, produced by MIRAGE-Large (frozen encoder, ConvNeXt decoder) fine-tuned only on GOALS layer annotations (background, nerve fibre layer, ganglion cell and inner plexiform layers, choroid). The anatomy-shaped and coverage strategies use precomputed soft guides from MIRAGE with frozen residual adapters. The anatomy-v1 guide uses a decoder-feature adapter; anatomy-v2 and COVER use an encoder-feature adapter. These guide products are not independently controlled in the cross-strategy comparison. CENTROID uses image intensity rather than a segmentation model: in the trained configuration its band (Equation 5) is 7 rows high and spans the central 10 columns of the patch grid (region fraction 0.28, lateral fraction 0.6). The adapters optimize feature-similarity objectives on FairVision-derived image caches, not a glaucoma classification loss, and are frozen during I-JEPA training. The retained provenance does not independently establish subject-level non-overlap for all data used to train the guidance models.

Included and excluded continuations. The early anatomy-v1 configuration resumes an ENVELOPE epoch-27 checkpoint rather than a documented epoch-25 fork, and its producing launch record is incomplete. Its only result is at epoch 30, where it lies $+0.0044$ above ENVELOPE (CI $[+0.0009, +0.0078]$). This comparison inherits the lineage uncertainty. Anatomy-v2 results use the continuation with full-precision target-encoder outputs. Probes from its earlier mixed-target-precision continuation are excluded, as are earlier uniform controls that used different target precision and context selection. The main table retains the comparable strategy endpoints, including the adverse COVER trajectory.

Mask measurement. Table 3 replays the sampler code in each strategy's trained configuration (commit `86ae8823f3afc9fa2c9ed6a3cb0c459b513ba1c5`), including the CENTROID band above and ENVELOPE placement on the hard envelope with the fallback rule used in training. The scope is 256 training volumes with 25 views each; every strategy sees the same crops, block sizes and placement seeds. Tissue is guide occupancy of at least 0.25 on the soft MIRAGE envelope, on the 16×16 patch grid. The review version used 600 B-scans from 24 volumes with seed 42, sizes drawn separately for each strategy, the code-default CENTROID band width and ENVELOPE placed on the soft guide. Table 6 compares both. In Table 3, "tissue hidden" is the share of tissue cells in the target union, "purity" the share of target cells on tissue, "unique targets" the number of unique target cells, "loss slots" the supervised target tokens including duplicates, and "context" the share of the grid supplied to the context encoder; in Table 6, "mask ratio" is the number of unique target cells divided by N. Repeated target indices add loss slots, and the anatomy-shaped and coverage strategies use different guide products from the ENVELOPE run. The band width has little effect on delivered context; most of the difference comes from replay noise and pairing, and for ENVELOPE also from the guide product and fallback rule.

Data use and ethics. This is secondary research on the public, de-identified FairVision release under its CC BY-NC-ND 4.0 data licence. No new participants were recruited or identifying fields added. A separate institutional ethics or exemption determination is not recorded for this project. The models are not clinically validated or intended for patient care. The illustrative OCT B-scan in Figures 1 and 2 comes from the separately attributed CC BY 4.0 OCTDL publication, not from the FairVision evaluation cohort.

Table 6: Review-version versus current mask geometry. Each cell gives the review-version value, then the value under the trained configuration and the protocol of Table 3. Per-image columns are measured one view at a time (6,400 single-view draws per seed), before batching; the last column is the context delivered at batch size 64.

	per image					delivered
strategy	tissue hidden	purity	mask ratio	context	loss slots	context
RANDOM	54.0 / 53.0	31.5 / 30.9	44.5 / 44.0	41.9 / 42.1	159.9 / 159.3	24.7 / 27.5
CENTROID	62.1 / 59.7	40.0 / 38.3	40.3 / 39.9	45.5 / 45.7	159.0 / 159.3	32.9 / 31.5
ENVELOPE	77.6 / 70.5	43.3 / 42.1	46.5 / 42.9	40.6 / 43.7	159.7 / 159.3	30.7 / 29.2
COVER	73.5 / 73.2	44.2 / 40.1	43.2 / 46.7	43.2 / 39.9	159.1 / 159.3	30.1 / 25.2
ANATOMY-V2	79.9 / 79.7	97.1 / 96.8	21.3 / 21.1	67.7 / 68.0	64.0 / 64.0	63.5 / 63.3

B Registered replication protocol

The replication plan was committed to the project repository before any new continuation produced a checkpoint and before any new probe (commit `86ae8823f3afc9fa2c9ed6a3cb0c459b513ba1c5`). Two amendments, committed before any new probe result, added a third seed index (commit `40d48f168ad8a04af47708eb08d2269be3a7e1c9`) and descriptive labels for the matched-budget control (commit `35554f92dfd4b0dc8a55c28bc71f49b8243a47e7`). RANDOM, CENTROID and ENVELOPE each continue from the shared epoch-25 checkpoint with three new seeds (the third added by a registered amendment), which set data order, crops, mask draws and data-loader streams. Runs with the same seed share data order only. ENVELOPE keeps the original run's fallback when no admissible window has low overlap: a uniform choice among admissible windows. A uniform-placement control receives CENTROID's per-image budgets with the first seed. Optimizer, schedule horizon and effective batch size match the original continuations, with full-precision teacher targets as in the originals; each run stops after its epoch-50 checkpoint. The original RANDOM and CENTROID continuations ran on different GPU hardware from the original ENVELOPE continuation and all new runs. The probe keeps the main protocol but runs in fp32, seeds each of five linear heads after the feature caches are created, and reports the mean AUC over head seeds together with their s.d.; the original encoders are re-probed in the same way. Test predictions were stored sealed (hashed, unscored) until the results freeze on 22 October 2026 and scored once afterwards; decisions during the runs used validation AUC only. For each new seed the primary contrasts are CENTROID and ENVELOPE minus RANDOM, with paired test-case bootstrap intervals. A seed index enters the primary analysis only if its RANDOM, CENTROID and ENVELOPE runs all finished before the results freeze (one did, so no label is assigned). With at least two such indices, a guided strategy is labelled consistent in direction only if every new-seed difference is positive and their mean exceeds the range of the RANDOM realizations (the original and the new runs of those indices). The matched control is described by the placement difference P (CENTROID minus RANDOM-CB) and the budget shift B (RANDOM-CB minus RANDOM) at the first seed, judged against the same range: placement difference not detected if the absolute value of P is within it, reversed if P is below its negative, placement accounting for most of the gap if P exceeds it and is at least half of P plus B, and both contributing otherwise. A positive mean otherwise counts as within run-to-run variation, and a non-positive mean as reversed. Analyses including the original runs, a visual-field mean-deviation regression within glaucoma cases, validation-split differences and pooling variants are secondary. Every started run is reported.

Table 7 lists every evaluated encoder with its paired interval and Table 8 the pooling variants, and Figure 4 shows the spread across runs. In glaucoma cases of the test split ($n = 1{,}466$), ridge regression on the frozen features predicts visual-field mean deviation with a mean absolute error from 3.44 to 3.55 dB across encoders, against 4.65 dB for the training mean (Table 9). Mean deviation partly defines the glaucoma label, so this is a severity endpoint on the same cohort, not an independent task.

C Guide agreement and envelope plausibility

FairVision has no retinal layer annotations, so neither guide can be scored for accuracy on it. We fixed failure rules before computing any rate and compared the CENTROID band with the hard MIRAGE envelope (patch occupancy at least 0.25) on full B-scans of 300 training volumes, 50 from

Table 7: Every evaluated encoder at epoch 50. AUC is the mean over five probe-head seeds, with their s.d. Δ is the difference from RANDOM of the same seed with a paired, label-stratified test-case bootstrap interval (10,000 resamples), and the validation difference is given for comparison. Intervals describe test-case sampling, not retraining. Original encoders are compared descriptively. Registered: in the primary analysis. Descriptive: seed index incomplete or run finished after the freeze.

policy	run	test AUC	head s.d.	val. AUC	Δ test (interval)	Δ val.	analysis
RANDOM	original	0.8641	0.0005	0.8467	—	—	original
RANDOM	seed A	0.8624	0.0005	0.8495	—	—	registered
RANDOM	seed B	0.8635	0.0008	0.8443	—	—	descriptive
CENTROID	original	0.8752	0.0006	0.8560	$+0.0111$	$+0.0093$	original
CENTROID	seed A	0.8721	0.0007	0.8547	$+0.0096$ $[+0.0052, +0.0142]$	$+0.0052$	registered
CENTROID	seed B	0.8754	0.0005	0.8566	$+0.0120$ $[+0.0077, +0.0164]$	$+0.0123$	descriptive
ENVELOPE	original	0.8755	0.0005	0.8597	$+0.0114$	$+0.0130$	original
ENVELOPE	seed A	0.8716	0.0002	0.8525	$+0.0092$ $[+0.0039, +0.0144]$	$+0.0030$	registered
ENVELOPE	seed B	0.8710	0.0011	0.8593	$+0.0075$ $[+0.0027, +0.0124]$	$+0.0150$	descriptive
RANDOM-CB	seed A	0.8650	0.0004	0.8463	$+0.0026$ $[-0.0015, +0.0066]$	-0.0032	control

Table 8: Pooling variants at epoch 50 (secondary, descriptive), given as pooling of patch tokens within a B-scan, then across B-scans: mean test AUC over the new seeds in the registered analysis, mean paired difference from RANDOM, and the label from the registered rule (consistent: consistent direction, within: within run-to-run variation). Runs lacking a variant are omitted for that variant.

pooling	RANDOM	CENTROID	ENVELOPE	Δ CENTROID	Δ ENVELOPE	label CENTROID	label ENVELOPE
mean, mean (primary)	0.8624	0.8721	0.8716	$+0.0096$	$+0.0092$	incomplete	incomplete
max, mean	0.8483	0.8625	0.8553	$+0.0142$	$+0.0069$	incomplete	incomplete
mean, max	0.8449	0.8637	0.8633	$+0.0188$	$+0.0184$	incomplete	incomplete
mean, mean and max	0.8553	0.8709	0.8719	$+0.0156$	$+0.0166$	incomplete	incomplete

each combination of glaucoma label and tertile of a signal-to-noise proxy, with ten B-scans per volume (3,000 B-scans). The computation was repeated on all 6,000 training volumes. A B-scan counts as a gross disagreement when the band–envelope IoU is below 0.2 or more than half of the band cells contain no envelope. Overall rates are population-weighted. Subgroup rates are unweighted proportions within the stratified sample. Disagreement rates use B-scans with a usable envelope, and intervals are 95% volume-cluster bootstrap intervals.

The band disagreed grossly with the envelope in 4.8% of B-scans (3.7–5.9, all training volumes 5.3%) and was off-centre in at least three of ten columns in 0.9%. Disagreement was more frequent in glaucoma than in normal eyes (6.5% versus 2.6%) and in the lowest than in the highest signal tertile (7.5% versus 2.3%). The median IoU was 0.41. Because the envelope is typically about half as thick as the 7-row band, an IoU of this size is expected even for a centred band. Of the envelopes, 3.8% (2.7–5.0) raised at least one plausibility flag: an empty or invalid envelope, three or more components (1.5%), irregular thickness (2.5%) or implausible position (0.4%). These are agreement and plausibility statistics between two automatic guides, not anatomical accuracy. The envelope model's only quantitative validation is in-domain on GOALS (foreground Dice 0.925).

D Target sizing and visible context

Batch equalization. Predictor targets are shortened to the minimum target length in the micro-batch, and the loss applies a single mean reduction to the stacked prediction and target tensors, including averaging over images. COVER chooses placements using the anatomical mass inside full candidate rectangles before this equalization. Because target indices are ordered row-major, shortening removes the lower portion of each rectangle rather than a spatially random subset. Coverage at placement time can consequently differ from coverage at the loss.

Measured geometry. Across 24,000 emitted targets, 73.4% remain perfect rectangles. In the batch-level audit of the trained configurations, COVER hides 73.3% of tissue cells against 70.7% for ENVELOPE while supplying less context (Table 3), and the guide mass hidden at the loss is 74.57% against 78.51% at placement (Table 10). Thus the trained COVER policy is not a controlled higher-coverage counterpart to ENVELOPE.

Table 9: Visual-field mean deviation (MD, dB) regressed from frozen epoch-50 features in glaucoma cases of the test split ($n = 1{,}466$), ridge regression with the penalty chosen on validation. Lower MAE and RMSE and higher Spearman correlation are better. Δ is the difference from RANDOM of the same seed with a paired case-bootstrap interval. MD partly defines the glaucoma label, so this is a severity endpoint on the same cohort, not an independent task. Encoders without a per-case feature-cache identity (RANDOM original) are not included.

policy	run	MAE	RMSE	Spearman	Δ MAE (interval)	Δ Spearman
RANDOM	seed A	3.55	4.72	0.52	—	—
RANDOM	seed B	3.49	4.64	0.53	—	—
CENTROID	original	3.45	4.57	0.53	—	—
CENTROID	seed A	3.47	4.60	0.53	-0.08 $[-0.16, -0.00]$	$+0.01$
CENTROID	seed B	3.46	4.58	0.54	-0.03 $[-0.10, +0.05]$	$+0.01$
ENVELOPE	original	3.46	4.60	0.54	—	—
ENVELOPE	seed A	3.50	4.64	0.54	-0.05 $[-0.13, +0.03]$	$+0.02$
ENVELOPE	seed B	3.44	4.60	0.53	—	—
RANDOM-CB	seed A	3.49	4.67	0.52	-0.06 $[-0.13, +0.01]$	-0.00
training-mean baseline		4.65	5.96	—		

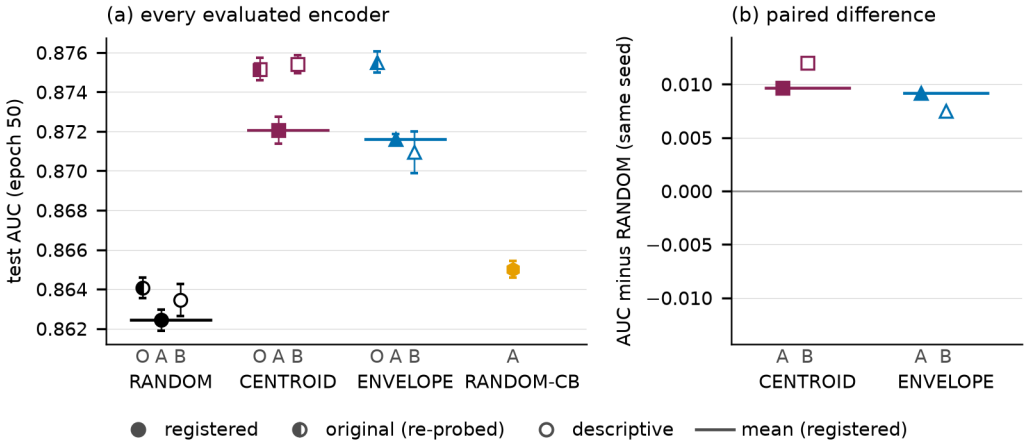


Figure 4: (a) Test AUC at epoch 50 of every evaluated encoder, averaged over five probe-head seeds (error bars: head-seed s.d.); bars are means over the registered seed indices. (b) Difference from RANDOM of the same seed index for each guided strategy, with its mean over the registered seed indices; no band is drawn because the registered range of RANDOM's realizations is not available. Letters give the seed index (O: original run, re-probed). Filled markers are runs in the registered analysis, half-filled markers re-probed original runs and open markers descriptive runs. Colours and marker shapes per strategy are those of the other figures. Continuations share an ancestor checkpoint, so the spread is across continuations, not independent initializations.

ENVELOPE shares the same shortening rule but does not optimize an exact coverage level. The anatomy-shaped strategies instead resample targets to a fixed $K{=}16$ indices. Equalization therefore has different consequences across the strategies, even when nominal target counts agree.

E Final target and context masks

Scope. We replayed the actual guided dataset, paired transforms, DataLoader worker collator and returned mask tensors on the existing 600-view training scope (24 volumes, 25 views each), using fixed new crop draws shared across policies. These are not reconstructions of the original training crops. Drawn block sizes were injected independently of placement; equal seeds alone do not pair policy-dependent rejection sampling. The production-size subset contains 576 views in complete batches of 64, with 573 valid post-crop guides. The shorter final batch is excluded from Table 10. Views within volumes are correlated, so these are descriptive mask statistics, not independent training replications.

What is measured. Target coverage is measured both at scoring and after target collation. Encoder-visible tissue is counted only in the final encoder indices, not in the full-grid target complement. Tissue is a guide occupancy thresholded at 0.25, not clinical ground truth. The context criterion is at least $\max(\lceil 0.21 N_{\text{tissue}} \rceil, \min(4, N_{\text{tissue}}))$ non-target tissue cells. This occupancy-cell diagnostic is distinct from the policy's soft-mass constraint on the target complement.

Table 10: Final masks from the trained COVER strategy on 576 fixed training views. Context-criterion misses count only the 573 valid guides. Scored and final coverage refer to supported soft guide mass; context counts include all views.

Measurement	COVER
Views losing tissue targets during equalization	300
Valid-guide context-criterion misses	358
Views with zero tissue context	55
Scored hidden guide mass (%)	78.51
Final hidden guide mass (%)	74.57
Mean encoder-visible tissue cells	10.13

Random fill is not a background quota. At full ramp all usable targets take the guided branch, while the `random_legal` fill mechanism is conditional on unused target blocks. Some views have no random-fill blocks. Such blocks can include tissue, and guided blocks can include background, so branch labels cannot stand in for background supervision. Target composition is therefore measured from the returned token indices rather than inferred from the branch name.

F Motivating attribution analysis and evidence limits

Origin of the target-selection question. Before studying anatomy-guided masking, we inspected feature-occlusion maps from glaucoma classifiers initialized with the uniformly masked I-JEPA baseline and then fine-tuned. The analysis compared mean pooling, cross-attention pooling and an attentive probe. At the B-scan level, a B-scan representation was zeroed before re-evaluating the classifier. At the patch level, an encoded patch was omitted from its B-scan's patch mean, the mean was recomputed from the remaining patches, and the altered B-scan representation was passed to the pooling module and classification head. The change in logit measures sensitivity to this feature intervention; it is not pixel occlusion or an attention map.

Let F denote the sequence of B-scan representations supplied to the classifier and $\ell(F)$ its glaucoma logit. If $F^{(-s,p)}$ replaces the representation of B-scan s with the mean of its remaining patch features after omitting patch p, the attribution is

$$\Delta_{s,p} = \ell(F) - \ell\Big(F^{(-s,p)}\Big). \tag{6}$$

Positive values mean that removing the feature lowers the glaucoma logit; negative values mean that removal raises it. Both signs indicate prediction sensitivity. Selected archived displays are not reproduced because of the dataset licence.

The historical report and selected overlays prompted us to investigate retinal regions as prediction targets. Those examples were selected from correctly classified cases and informative B-scans, so they are not a population-level estimate of anatomical localization. Nor does a response to feature removal establish a clinically causal region or show which pretraining mask is best. Fine-tuning used diagnosis labels, so the maps cannot separate the contribution of self-supervised pretraining from subsequent supervised adaptation.

Available evidence. Archived occlusion curves and heat maps remain available, but the underlying per-volume numerical arrays and a complete normalization record are not retained in the evidence used for this paper. The historical record also documents sensitivity of patch-level displays to numerical precision and colour scaling. The selected image panels are retained as archived visualizations, without restoring their numerical summaries as validated results. We make no spatial-consistency, laterality or clinical landmark claim from those displays. Recovering the producing arrays, checkpoint identities and plotting transforms would be necessary for a reproducible quantitative analysis. No new attribution campaign was run, and the saved frozen-probe AUC comparisons are unchanged.

Table 11: Original-run contrasts: matched epoch *and* matched precision. Δ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and p is a DeLong test for correlated ROC curves. q is Benjamini–Hochberg over these nine contrasts. All p and q values here are descriptive (Section 6). Every comparison against the null excludes zero.

contrast	epoch	Δ AUC	95% CI	p	q
ENVELOPE − RANDOM	50	+0.0120	[+0.0068, +0.0173]	<0.0001	<0.0001
ENVELOPE − RANDOM	75	+0.0080	[+0.0028, +0.0132]	0.0025	0.0045
ENVELOPE − RANDOM	100	+0.0062	[+0.0010, +0.0113]	0.0199	0.0299
CENTROID − RANDOM	50	+0.0099	[+0.0051, +0.0149]	<0.0001	0.0001
CENTROID − RANDOM	75	+0.0113	[+0.0063, +0.0164]	<0.0001	<0.0001
CENTROID − RANDOM	100	+0.0109	[+0.0057, +0.0160]	<0.0001	<0.0001
CENTROID − ENVELOPE	50	-0.0020	[-0.0069, +0.0029]	0.4114	0.4114
CENTROID − ENVELOPE	75	+0.0033	[-0.0013, +0.0079]	0.1491	0.1677
CENTROID − ENVELOPE	100	+0.0047	[+0.0004, +0.0091]	0.0294	0.0378

G Full paired-contrast table

Benjamini–Hochberg is applied within a declared family consisting of exactly these nine contrasts (the fp16 comparisons among RANDOM, CENTROID and ENVELOPE at epochs 50, 75 and 100) and the remaining 25 exploratory contrasts in the stored statistics are corrected as a separate family. The two families are never pooled, and the q values printed above are those of the nine-contrast family only. Declaring the family fixes what is corrected together. It does not make these contrasts confirmatory, because the test split was inspected repeatedly before the family was fixed, and the paper reports them as descriptive throughout.

H Probe precision control

The original rectangle-strategy probes use mixed precision, while the anatomy-shaped and coverage-strategy probes use fp32. All Δ values in Table 2 use a baseline matched in both epoch and probe precision. Probe precision is distinct from pretraining target-encoder precision. We repeated the recorded mixed-precision probes with autocast disabled in feature extraction, head fitting and evaluation. Each comparison uses the same frozen encoder. The recorded changes are smaller than the principal strategy differences.

Table 12: Full fp32 re-probe against the original fp16 result, same frozen encoder in each row. The eight DeLong p-values are unadjusted diagnostics, not a family-wise test. The largest difference is 2×10^{-4}.

policy	epoch	fp16 AUC	fp32 AUC	Δ	p
ENVELOPE	50	0.876064	0.876063	-0.000001	0.964
ENVELOPE	75	0.880307	0.880305	-0.000002	0.806
ENVELOPE	100	0.880743	0.880761	+0.000018	0.167
CENTROID	50	0.874030	0.874015	-0.000015	0.412
CENTROID	100	0.885485	0.885293	-0.000192	0.767
RANDOM	50	0.864097	0.864121	+0.000024	0.128
RANDOM	75	0.872302	0.872302	-0.000001	0.962
RANDOM	100	0.874581	0.874485	-0.000096	<0.001